

Yu Hao

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My research focuses on operating-system security and program analysis for C/C++ systems, with recent work on applying LLMs to security analysis and software engineering tasks. I build and evaluate tools that combine fuzzing, static analysis, symbolic execution, and LLM-based program analysis for real-world systems such as Linux, Android, and HarmonyOS kernels. My work includes LLM-assisted static analysis, automated kernel fuzzing specification generation, and exploitability/security analysis workflows. These projects have led to papers in S&P, USENIX Security, CCS, NDSS, ICSE, FSE, OOPSLA, and TSE, several open-source tools, invited talks, Linux kernel bug/CVE reports, and patches.

Education

- 09/2018 – 09/2024 **Ph.D.** in Computer Science, University of California, Riverside, USA.
Advisor: [Zhiyun Qian](#)
◆ Operating System Security, LLMs for Security, Fuzzing, Symbolic Execution, Static Analysis
- 09/2015 – 06/2018 **Master** in Control Science and Engineering, Xi'an Jiaotong University, China.
Advisor: Ting Liu
◆ Concurrency Security, Symbolic Execution
- 09/2011 – 06/2015 **Bachelor** in Control Science and Engineering, Xi'an Jiaotong University, China.
- 09/2008 – 06/2011 Hohhot No.2 High School, Inner Mongolia, China.

Experience

- 10/2024 – present **Security Researcher**
HarmonyOS System Security Team, Huawei, China
◆ HarmonyOS Security, Fuzzing, LLMs for Security, Static Analysis
- 06/2023 – 09/2023 **Research Intern**, Mentor: [Weidong Cui](#)
Systems Security and Privacy Research Group, Microsoft Research, USA
◆ LLMs for Security / Program Analysis
- 06/2022 – 09/2022 **Research Intern**, Mentor: Kris Alder
Android Security Team, Google, USA
◆ Linux/Android Kernel Security, Fuzzing, Static Analysis
- 09/2018 – 09/2024 **Graduate Student Researcher**, Advisor: [Zhiyun Qian](#)
Center for Research and Education in Cyber Security and Privacy (CRESP), University of California, Riverside, USA
◆ Operating System Security, LLMs for Security, Fuzzing, Symbolic Execution, Static Analysis
- 06/2013 – 06/2018 **Research Assistant**, Advisor: Ting Liu
Ministry of Education Key Lab For Intelligent Networks and Network Security, Xi'an Jiaotong University, China
◆ Concurrency Security, Symbolic Execution

Thesis

- [PhD] Expanding the Boundaries of OS Kernel Fuzzing [📄 Permalink](#)
• [C9, ICSE 22, Google Research Paper Rewards][C10, S&P 23][C17, CCS 25, Distinguished Paper Award]
- [Master] Research on Malicious Code Hiding Methods Based on Uncertain Interleaving of Concurrent Programs [📄](#)
• [C3, QRS 21, Best Paper Award][J3, *IEEE Transactions on Reliability*]

Publications

- [C]: Conference, [J]: Journal, [P]: Preprint
- [C18] NeuroMerge: ML-Guided State Merging for Efficient Symbolic Execution
- ISSRE 26 Shenghan Zhng, Shitong Zhu, Yu Hao, Xingyu Li, Keyu Man, Zhang Zheng, Qing Deng, Zhiyun Qian, Srikanth Krishnamurthy
IEEE International Symposium on Software Reliability Engineering, ISSRE 26.

- [C17]
CCS 25 SyzSpec: Specification Generation for Linux Kernel Fuzzing via Under-Constrained Symbolic Execution
Yu Hao, Juefei Pu, Xingyu Li, Zhiyun Qian, Ardalan Amiri Sani
ACM SIGSAC Conference on Computer and Communications Security, CCS 25.   Tool
• [Distinguished Paper Award]
- [C16]
S&P 25 SCAD: Towards a Universal and Automated Network Side-Channel Vulnerability Detection
Keyu Man, Zhongjie Wang, Yu Hao, Shenghan Zheng, Xin'an Zhou, Yue Cao, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 25.   Tool
- [C15]
USENIX Security 24 SymBisect: Accurate Bisection for Fuzzer-Exposed Vulnerabilities
Zheng Zhang, Yu Hao, Weiteng Chen, Xiaochen Zou, Xingyu Li, Haonan Li, Yizhuo Zhai, Zhiyun Qian, Billy Lau
USENIX Security Symposium 2024.   Tool 
• [Linux Security Summit 25]
- [C14]
OOPSLA 24 Enhancing Static Analysis for Practical Bug Detection: An LLM-Integrated Approach
Haonan Li, Yu Hao, Yizhuo Zhai, Zhiyun Qian
ACM SIGPLAN International Conference on Object-Oriented Programming Systems, Languages, and Applications, OOPSLA 24.   Tool 
- [C13]
S&P 24 SyzGen++: Dependency Inference for Augmenting Kernel Driver Fuzzing
Weiteng Chen, Yu Hao, Zheng Zhang, Xiaochen Zou, Dhilung Kirat, Shachee Mishra, Douglas Schales, Jiyong Jang, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 24.   Tool 
- [C12]
NDSS 24 SyzBridge: Bridging the Gap in Exploitability Assessment of Linux Kernel Bugs in the Linux Ecosystem
Xiaochen Zou, Yu Hao, Zheng Zhang, Juefei Pu, Weiteng Chen, Zhiyun Qian
Network and Distributed System Security Symposium, NDSS 24.   Tool 
- [P1]
E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
Yu Hao, Weiteng Chen, Ziqiao Zhou, Weidong Cui 
• [AGI Leap Summit 2024] [Symposium on the Science of Security 24]
- [C11]
FSE 23 Assisting Static Analysis with Large Language Models: A ChatGPT Experiment
IVR Haonan Li, Yu Hao, Yizhuo Zhai, Zhiyun Qian
The ACM International Conference on the Foundations of Software Engineering, Ideas, Visions and Reflections, FSE 23 IVR.   Tool  
- [C10]
S&P 23 SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
Yu Hao, Guoren Li, Xiaochen Zou, Weiteng Chen, Shitong Zhu, Zhiyun Qian, Ardalan Amiri Sani
IEEE Symposium on Security and Privacy, S&P 23.   Tool  Result 
• [Linux Security Summit 23] [Qualcomm Product Security Summit 23]
• [Symposium on the Science of Security 24]
- [C9]
ICSE 22 Demystifying the Dependency Challenge in Kernel Fuzzing
Yu Hao, Hang Zhang, Guoren Li, Xingyun Du, Zhiyun Qian, Ardalan Amiri Sani
IEEE/ACM International Conference on Software Engineering, ICSE 22.   Tool  
• [Google Research Paper Rewards]
- [C8]
NDSS 22 Progressive Scrutiny: Incremental Detection of UBI bugs in the Linux Kernel
Yizhuo Zhai, Yu Hao, Zheng Zhang, Weiteng Chen, Guoren Li, Zhiyun Qian, Chengyu Song, Manu Sridharan, Srikanth V. Krishnamurthy, Trent Jaeger, Paul Yu
Network and Distributed System Security Symposium, NDSS 22.   Tool 
• [2023 Cyber Security CRA Capstone Poster]
- [C7]
ACSAC 21 Eluding ML-based Adblockers With Actionable Adversarial Examples
Shitong Zhu, Zhongjie Wang, Xun Chen, Shasha Li, Keyu Man, Umar Iqbal, Zhiyun Qian, Kevin S Chan, Srikanth V Krishnamurthy, Zubair Shafiq, Yu Hao, Guoren Li, Zheng Zhang, Xiaochen Zou
Annual Computer Security Applications Conference, ACSAC 21.   Tool 
- [C6]
CCS 21 Themis: Ambiguity-Aware Network Intrusion Detection based on Symbolic Model Comparison
Zhongjie Wang, Shitong Zhu, Keyu Man, Pengxiong Zhu, Yu Hao, Zhiyun Qian, Srikanth V. Krishnamurthy, Tom La Porta, Michael J. De Lucia
ACM SIGSAC Conference on Computer and Communications Security, CCS 21.   Tool 
- [C5]
CCS 21 Statically Discovering High-Order Taint Style Vulnerabilities in OS Kernels
Hang Zhang, Weiteng Chen, Yu Hao, Guoren Li, Yizhuo Zhai, Xiaochen Zou, Zhiyun Qian
ACM SIGSAC Conference on Computer and Communications Security, CCS 21.   Tool 

- [C4]
FSE 20 UBITect: A Precise and Scalable Method to Detect Use-before-Initialization Bugs in Linux Kernel
Yizhuo Zhai, Yu Hao, Hang Zhang, Daimeng Wang, Chengyu Song, Zhiyun Qian, Mohsen Lesani, Srikanth V. Krishnamurthy, Paul Yu
ACM SIGSOFT International Symposium on Foundations of Software Engineering, FSE 20.   
• [2023 Cyber Security CRA Capstone Poster]
- [J3]
ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, Yu Hao, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE Transactions on Reliability.   
- [C3]
QRS 21 ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, Yu Hao, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE International Conference on Software Quality, Reliability, and Security, QRS 21.  
• [Best Paper Award]
- [J2]
TSE Tell You a Definite Answer: Whether Your Data is Tainted During Thread Scheduling
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Yu Hao, Pei Liu, Ting Liu
IEEE Transactions on Software Engineering, TSE.    
• [S&P 17 Poster] Patent: [PCT] [CN]
- [J1]
Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Yu Hao, Pei Liu, Lechen Yu, Ting Liu
IEEE Access.  
- [C2]
ICST 17 Automated Testing of Definition-Use Data Flow for Multithreaded Programs
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Pei Liu, Jialiang Chang, Yu Hao, Ting Liu
IEEE International Conference on Software Testing, Verification and Validation, ICST 17.  
- [C1]
SATE 17 Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Yu Hao, Pei Liu, Lechen Yu, Ming Fan, Ting Liu
IEEE International Conference on Software Analysis, Testing and Evolution, SATE 16.  

Patent

WO2017181628 - TAINT ANALYSIS METHOD EMPLOYING SYMBOLIC COMPUTATION AND USED FOR DYNAMIC PARALLEL PROGRAM [PCT] [CN]

Presentations, Talks

- E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
 - AGI Leap Summit 2024  Website
 - Symposium on the Science of Security, HoTSoS 2024  Website
- SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
 - Linux Security Summit North America 2023  Website  Slides  Recording
 - Qualcomm Product Security Summit 2023  Website
 - 44th IEEE Symposium on Security and Privacy, S&P 2023  Website  Recording
 - Symposium on the Science of Security, HoTSoS 2024  Website
- Demystifying the Dependency Challenge in Kernel Fuzzing
 - 44rd IEEE/ACM International Conference on Software Engineering, ICSE 2022  Website  Slides  Recording

Honors, Awards

- 2025 CCS 2025, Distinguished Paper Award
- 2024 Laxmi Bhuyan Fellowship Award, University of California, Riverside
Dissertation Completion Fellowship Award, University of California, Riverside
- 2023 IEEE S&P Student Travel Grant
Google Research Paper Rewards
The Linux Foundation's Travel Fund
- 2021 QRS 2021 Best Paper Award
- 2018 Dean's Distinguished Fellowship, University of California, Riverside
Excellent Graduated Student, Xi'an JiaoTong University
- 2017 Excellent Graduate Student, Xi'an JiaoTong University

	Graduate Student Fellowship, Xi'an JiaoTong University
	Sheng Han High IQ Association
	IQ ≥ 135 , Club at the same level as Mensa
2016	Graduate Student Fellowship, Xi'an JiaoTong University
2015	Graduate Student Fellowship, Xi'an JiaoTong University
2014	Siyuan Scholarship, Xi'an JiaoTong University
2013	Siyuan Scholarship, Xi'an JiaoTong University
	Second Prize, Xi'an JiaoTong University ACM-ICPC Competition
	Second Prize, Xi'an JiaoTong University MCM Competition
2010	First Prize, Mathematical Olympiad in Provinces, Inner Mongolia
	Ranking 4th at that year in the province with population of 25 million
	Third Prize, Physics Olympiad in Provinces, Inner Mongolia
	Third Prize, Chemistry Olympiad in Provinces, Inner Mongolia
2007	First Prize, National Mathematics Invitational Competition, Inner Mongolia
	A total of 4 at that year in the province with population of 25 million

Academic Service

Program Committee	CCS 2026. RAID 2026. FORGE@ICSE 2025, 2024. MSR 2024(Junior PC)
Reviewer	TIFS 2024, 2023. TDSC 2026. TOSEM 2026, 2025. Computing Surveys 2026, EAI SecureComm 2023
Sub-Reviewer	S&P 2025, 2024, 2021, 2020. CCS 2024. USENIX Security 2025, 2021. NDSS 2025, 2021, 2020
AEC	EuroSys 2023. ISSTA 2024. ECOOP 2024

References

Advisor of Ph.D. Degree	Zhiyun Qian <i>Everett and Imogene Ross Professor</i> Computer Science and Engineering department, University of California, Riverside, USA Contact: zhiyunq@cs.ucr.edu
Collaborator	Ardalan Amiri Sani <i>Associate Professor</i> Computer Science Department, University of California, Irvine, USA Contact: ardalan@uci.edu
Collaborator	Srikanth V. Krishnamurthy <i>Professor, IEEE Fellow</i> Computer Science and Engineering department, University of California, Riverside, USA Contact: krish@cs.ucr.edu
Intern Mentor at Microsoft Research	Weidong Cui <i>Partner Research Manager</i> Systems Security and Privacy Research Group, Microsoft Research, Redmond, USA Contact: Weidong.Cui@microsoft.com
Advisor of Master Degree	Ting Liu <i>Professor, Associate Dean</i> School of Cyber Science and Engineering, Xi'an Jiaotong University, China Contact: tingliu@mail.xjtu.edu.cn